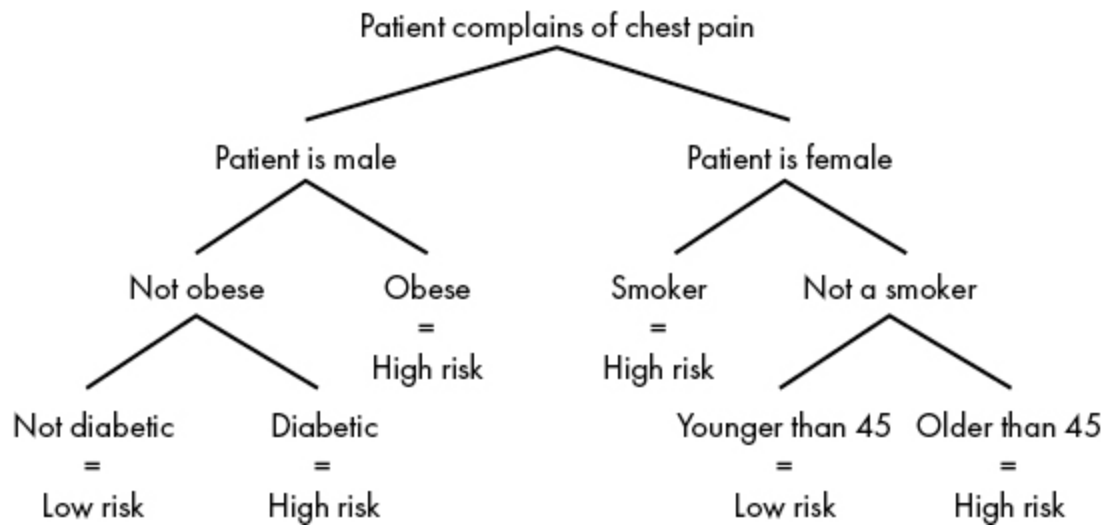




[Figure 9-1](#): A simplified decision tree for heart attack triage

You can read this diagram from top to bottom. At the top, we can see that the heart-attack diagnosis process begins with a patient reporting chest pain. After that, the process branches out depending on the sex of the patient. If the patient is a man, the diagnosis process continues in the left branch and we determine whether he is obese. If the patient is a woman, the process continues in the right branch instead, and we determine whether she is a smoker. At each point in the process, we follow the appropriate branch until we reach the bottom of the tree, where we find the tree's classification of whether the patient is at high risk or low risk for a heart attack. This binary branching process resembles a tree whose trunk branches into smaller offshoots until reaching the ends of the farthest branches. Accordingly, the decision process illustrated in [Figure 9-1](#) is called a decision tree.

Every place you see text in [Figure 9-1](#) is a *node* of the decision tree. A node like “Not obese” is known as a *branching node* because there's at least one more branch to follow before we're able to make a prediction. The “Not diabetic = Low risk” node is a *terminal node* because if we've arrived there, we don't need to branch anymore and we know the decision tree's final classification (“Low risk”).

If we could design a thorough, well-researched decision tree that always led to good triage decisions, it's possible that someone without medical training could perform triage of heart attack patients, which would save every emergency room in the world plenty of money because they would no